

We use the integers ***a***, ***b***, and ***n*** to create the following series:

Problem

$(a + 2^0 \cdot b), (a + 2^0 \cdot b + 2^1 \cdot b), \dots, (a + 2^0 \cdot b + 2^1 \cdot b +$

You are given ***q*** queries in the form of ***a***, ***b***, and ***n***. For each query, print the series corresponding to the given ***a***, ***b***, and ***n*** values as a single line of ***n*** space-separated integers.

Input Format

The first line contains an integer, ***q***, denoting the number of queries.

Each line ***i*** of the ***q*** subsequent lines contains three space-separated integers describing the respective ***a_i***, ***b_i***, and ***n_i*** values for that query.

Constraints

- $0 \leq q \leq 500$
- $0 \leq a, b \leq 50$
- $1 \leq n \leq 15$

Output Format

For each query, print the corresponding series on a new line. Each series must be printed in order as a single line of ***n*** space-separated integers.

Sample Input

```
2
0 2 10
5 3 5
```

Sample Output

```
2 6 14 30 62 126 254 510 1022 2046
8 14 26 50 98
```

Explanation

We have two queries:

1. We use ***a* = 0**, ***b* = 2**, and ***n* = 10** to produce some series

s₀, ***s₁***, ..., ***s_{n-1}***:

- ***s₀* = 0 + 1 · 2 = 2**
- ***s₁* = 0 + 1 · 2 + 2 · 2 = 6**
- ***s₂* = 0 + 1 · 2 + 2 · 2 + 4 · 2 = 14**

... and so on.

Once we hit ***n* = 10**, we print the first ten terms as a single line of space-separated integers.

2. We use ***a* = 5**, ***b* = 3**, and ***n* = 5** to produce some series

s₀, ***s₁***, ..., ***s_{n-1}***:

```

16      arr1[i][1] = a;
17      arr1[i][2] = n;
18      }
19
20      int num = 0;
21      for(int i = 0; i < t ; i++) {
22          //Calculating each query
23
24          for (int j = 0; j<arr1[i][2];j++) {
25              num = num + (int) Math.pow(2, j);
26              System.out.print((arr1[i][0]+num * arr1[i][1])+" ");
27          }
28
29          System.out.println();
30          num = 0;    //Resetting the value of num for next query
31      }
32      }
33      }
34
35
```

Line: 33 Col: 1

☐ Test against custom input



You have earned 10.00 points!

You are now 23.33 points away from the 2nd star for your java badge.

7%

26.67/50

Congratulations

You solved this challenge. Would you like to challenge your friends?

Test case 0

Compiler Message

Test case 1

Success

Test case 2

Input (stdin)

```
1 2
2 0 2 10
3 5 3 5
```

Test case 3

Test case 4

Expected Output

```
1 2 6 14 30 62 126 254 510 1022 2046
2 8 14 26 50 98
```